What are Vector, Vector Embedding

aalo
pyaz
lady finger

Amazon

first thought principle

Recommendation system

Youtube search system

1000 web development course

DB

[]

vector

Rohit.js

[10,12,18,9,2,91,110,18]

javscript 1

web development course

web development full course

5-6 chejin dikha de raha

muje javascript ki kitab dede

10 20 30 40 50

javacript book

First Solution?

product

1 Million

100 product

| | | | | |
|---|---|---|---|---|
| Protein | BCAA | Creatine | Shaker | Yoga Mat |
| Apple | Orange | Banana | Mango | WaterMelon |
| Tomato | Potato | onion | baigan | Lady Finger |
| pant | Shirt | Shoes | Belt | Shocks |
| Camera | Mic | tripod | Ring light | earphone |

color

| | | | | |
|---|---|---|---|---|
| orange | red | green | pink | purple |

Problem:

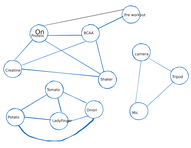
1: It's Not Scalable (The Manual Labor Problem)

2: The Boundaries are Too Rigid (The "Blender" Problem) or Banana with protein

3: It Can't Discover New Relationships

Beer and diaper story??

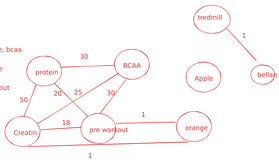
Second Solution--> Graph based solution




1: protein, creatine, bcaa

2: Protein , creatine

3: creatine, prworkout
orange




1: New user

protein

creatine, bcaa, preworkout

Myprotein

1 Million Product

remote

creatine mic

top: 5

| | Protein | Bcaa | Creatine | Mic | Camera |
|---|---|---|---|---|---|
| Protein | 0 | 51 | 40 | 4 | 1 |
| BCAA | 51 | 0 | 35 | 3 | 1 |
| Creatine | 40 | 35 | 0 | 2 | 0 |
| Mic | 4 | 3 | 2 | 0 | 30 |
| Camera | 1 | 1 | 0 | 30 | 0 |

remote

Limitation:

1: Size of matrix will be N*N

2: Selecting the array and then sort it will be time consuming

3: The "Cold Start" Problem

4: It Only Captures Co-occurrence, Not Content(On Whey vs My Protein Whey)

Third Solution: Assigning number to each item and find closeness

we will assign number to each item and then recommend
other item which is closer to that item...

apple: 1
banana.: 2
watermelon: 3
Papaya: 4
guava: 5
Mango: 6
.
.
.
peach: 100

protein: 101
bcaa: 102
creatine: 103
preworkout 104
.
.
.
fish Oil: 150

Blender: 151
Toaster: 152
Coffer Maker: 153
Tea:200
coffe:201

consumable/Appliance
fitness/junk

Limitation:
1: The "Single Meaning" Problem
2: The "Boundary" Problem
3: The "Insertion" Problem

4: Vector is a solution:

[10,14,19,40,90]

Netflix Movie recommendation system

1: Dhamal
2: Hera Pehri

Movie

comedy
action

comedy
realLife

we have set of 20 movies, a person love one movie,
which movie should we recommend them....

dhamal

action,comedy,horror,real,emotinal

dhamal=[2,9]

Hera Pehri=[1,10]

X-axis: Action Level (-10 for Peaceful, +10 for Action)
Y-axis: Comedy Level (-10 for Serious, +10 for Comedy)

action comedy
war: [10, -5]

The "Serious-Drama" Blob (Bottom-Left): Lagaan, Swades, Dangal, and Taare Zameen Par are all clustered together. But are they really that similar?
Lagaan is a historical sports epic.
Swades is a contemporary social drama.
Dangal is a biographical sports drama.
Taare Zameen Par is an emotional film about childhood and disability.
A user who loved Lagaan for its historical scale might be disappointed by a recommendation for Taare Zameen Par. The 2D model thinks they are the same because they are both "not action" and "not comed

The "Comedy" Blob (Top-Left): Hera Pheri is a slapstick comedy.
3 Idiots is a social satire. Jab We Met is a romantic comedy.
The 2D model lumps them all together, failing to understand the crucial difference between "romance" and "slapstick".

vector

Upgrading to a 5-Dimensional Model

a: [10,5,7,-4,3]
b: [11,5,2,6,-4]

c: [12,6,7,-4,3]

Dim 1: Action Level (same as before)
Dim 2: Comedy Level (same as before)
Dim 3: Emotional Drama (-10 for Light-hearted, +10 for Intense/Tearjerker)
Dim 4: Romance Level (-10 for No Romance, +10 for Epic Love Story)
Dim 5: Realism (-10 for Pure Escapism/Fantasy, +10 for Grounded/Realistic)

| | | |
|---|---|---|
| War | [10, -5] | [ 10, -5, -4, 2, -7 ] |
| Pathaan | [10, -3] | [ 10, -3, -5, 3, -9 ] |
| Sholay | [8, 5] | [ 8, 5, 7, 6, 2 ] |

king [10,9,10,10,]
Men: [10,9,3,2,]

vector("King") - vector("Man") + vector("Woman") ≈ vector("Queen")

796, 4096,12k

How to measure the closeness between these vector

Euclidean Distance: Measures the straight-line "ruler" distance between two points.
It cares about the magnitude of the difference.

Cosine Distance: Measures the "angle" between two vectors.
It cares about the orientation or direction, not the magnitude.

Eucledian Formula:

For two vectors, $A = (a_1, a_2, \ldots, a_n)$ and $B = (b_1, b_2, \ldots, b_n)$, the Euclidean distance is:
$d(A, B) = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \ldots + (a_n - b_n)^2}$

Cosine Formula:
Cosine Similarity(A, B) = (A · B) / (||A|| * ||B||)

A · B is the dot product of the vectors ($a_1*b_1 + a_2*b_2 + \ldots$).
||A|| is the magnitude (or L2 norm) of vector A ($\sqrt{a_1^2 + a_2^2 + \ldots}$).

Value will be: [-1, 1]